AQAL Platform — Master Research Reference Guide

The Single Source of Truth for All Empirical Findings

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Total Verified Citations: 200+
Purpose: Consolidate every research finding, DOI, effect size, and verdict from all audit waves into one document. Use this to update the site with the most accurate information.

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1. Therapy Library — 143 Therapies Audited

Rating Distribution

Rating	Count
Strong	78
Moderate	57
Emerging	5
Debunked	3

Top 15 Strongest Interventions (by effect size)

#	Therapy	Effect Size	Citation	Target Lines
1	Mastery Experiences in Off-Job Time for recovery and wellbeing	Positive meta-analytic associations for mastery with job and personal outcomes; k = 316 independent samples, N = 99,329 participants	Headrick, L., Newman, D. A., Park, Y. A., & Liang, Y. (2023). Recovery experienc	Not mapped
2	Social Prescribing and Community Referral for loneliness	moderate reductions in loneliness, k = 26 studies	Wang, Y., Xu, Y., Gao, L., & Bai, X. (2026). Impact of social prescribing on qua	Not mapped
3	Equine-assisted therapy for trauma PTSD and emotional regulation	d = 12.46 (mean difference), 95% CI [9.03, 15.88], k = 13 studies	Provan, M., & Provan, C. (2024). Are equine-assisted services beneficial for mil	Emotional Regulation, Trauma Recovery, I
4	Stellate Ganglion Block	Adjusted mean CAPS-5 score change = -12.6 points (95% CI, -15.5 to -9.7 points) vs -6.1 points for sham (P = .01)	Rae Olmsted, K. L., Bartoszek, M., Mulvaney, S., McLean, B., Turabi, A., Young,	Physiological hyperarousal, psychologica
5	Transcendental Meditation ™	d = -4.7 mm Hg (95% CI [-7.4, -1.9]) for systolic BP, d = -3.2 mm Hg (95% CI [-5.4, -1.3]) for diastolic BP, k = 9 studies	Anderson, J. W., Liu, C., & Kryscio, R. J. (2008). Blood pressure response to tr	Cognitive, Emotional, Physical

#	Therapy	Effect Size	Citation	Target Lines
6	Internal Family Systems (IFS) therapy	d = -4.46 (CAPS) and d = -3.05 (DTS)	Hodgdon, H. B., Anderson, F. G., Southwell, E., Hrubec, W., & Schwartz, R. (2022	Trauma Integration, Emotional Regulation
7	Sauna and Heat Therapy for cardiovascular and mental health	MD = -3.94 mmHg, 95% CI [-7.22, -0.67], k = 10 studies	Pizzey, F. K., Smith, E. C., & Ruediger, S. L. (2021). The effect of heat therap	Not mapped
8	Sleep Hygiene Education for insomnia prevention	MD = 3.4, 95% CI [2.08, 4.64], k = 42 studies	Ruan, J. Y., Liu, Q., Chung, K. F., Ho, K. Y., & Yeung, W. F. (2025). Effects of	Not mapped
9	Transcranial magnetic stimulation (TMS)	OR = 3.38, 95% CI [2.24, 5.12], k = 81 studies	Brunoni, A. R., Chaimani, A., Moffa, A. H., et al. (2017). Repetitive transcrani	Emotional Regulation, Cognitive Flexibil
10	Psilocybin-assisted therapy	d = 2.5, 95% CI [1.4, 3.5], k = 1 study	Davis, A. K., Barrett, F. S., May, D. G., Cosimano, M. P., Sepeda, N. D., Johnso	Emotional Regulation, Meaning-Making, Co
11	Drum circles and rhythmic entrainment for social bonding and mood	d = -2.14 to -3.41 (depression reduction), d = 7.69 to 10.59 (social resilience increase), k = 1 RCT	Fancourt, D., Perkins, R., Ascenso, S., Carvalho, L. A., Steptoe, A., & Williamo	Social, Emotional, Interpersonal
12	Creative arts therapies (drama therapy and	SMD = -1.98, 95% CI [-3.8, -0.16], k = 7 studies	Wang, J., Zhang, B., Yahaya, R., & Abdullah, A. B.	Emotional, Interpersonal, Cognitive

#	Therapy	Effect Size	Citation	Target Lines
	psychodrama) for trauma		(2025). Colors of the mind: a	
13	Intermittent Fasting and Time-Restricted Eating	-1.8 kg (95% CI, -3.2 to -0.4), N = 139	Jamshed, H., et al. (2022). Effectiveness of Early Time-Restricted Eating for We	Not mapped
14	Progressive Muscle Relaxation for anxiety and insomnia	SMD = -1.74, 95% CI [-2.14, -1.34], k = 31 studies	Donato, K. O., Falcão, L., Nishizima, A., Oliveira, A. S., et al. (2026). Progre	Not mapped
15	Narrative Therapy	SMD = -1.64, 95% CI [-1.95, -1.32], k = 43 studies	Hu, G., Han, B., Gains, H., & Jia, Y. (2024). Effectiveness of narrative therapy	Identity Coherence, Meaning-Making, Emot

Complete Therapy List (All 143)

Strong (78 therapies)

Therapy	Effect Size	Primary Citation	Target Lines
Self-Compassion interventions (Kristin Neff framework)	g = 0.75 for self-compassion,	Ferrari, M., Hunt, C., Harrysunker, A., Abbott, M. J., Beath, A. P., &	Not mapped
Intermittent Fasting and Time-Restricted Eating	-1.8 kg (95% CI, -3.2 to -0.4)	Jamshed, H., et al. (2022). Effectiveness of Early Time-Restricted Eat	Not mapped
Nature Dose 120 minutes per week	OR = 1.59, 95% CI [1.31, 1.92]	White, M. P., Alcock, I., Grellier, J., Wheeler, B. W., Higgins, M., H	Not mapped
Dance therapy and recreational dance	MD = 1.58, 95% CI [0.21, 2.95]	Hewston, P., Kennedy, C. C., Borhan, S., Merom, D., Santaguida, P., Io	Not mapped
Forgiveness interventions	Δ+ = 0.56, 95% CI [0.43, 0.68]	Wade, N. G., Hoyt, W. T., Kidwell, J. E. M., & Worthington, E. L. (201	Not mapped
Clinical Hypnosis	d = 0.60, p ≤ .001, k = 40 pos	Milling, L. S., Valentine, K. E., LoStimolo, L. M., et al. (2021). Hyp	Not mapped
Psychodynamic Therapy	d = 0.97 (post-treatment), d =	Shedler, J. (2010). The efficacy of psychodynamic psychotherapy. Ameri	Not mapped
Bouldering and rock climbing for depression	d = 0.89 (short term), d = 1.1	Luttenberger, K., Karg-Hefner, N., Dorscht, L., et al. (2022). Boulder	Not mapped
Cyclic Sighing / Physiological Sigh	effect size = 0.736, N = 114	Balban, M. Y., Neri, E., Kogon, M. M., Weed, L., Nouriani, B., Jo, B.,	Not mapped

Therapy	Effect Size	Primary Citation	Target Lines
Yoga Nidra / Non-Sleep Deep Rest (NSDR)	Hedge’s g = -0.80 (active comp	Ghai, S., Odyniec, P., & Ghai, I. (2026). Effects of Yoga Nidra on Str	Not mapped
Cognitive Defusion techniques (ACT component) for anxiety and rumination	Hedges’ g = 1.04 (believabilit	Oppo, A., Pasquini, L., Misitano, A., Savoia, A., & Presti, G. (2026).	Not mapped
Self-Distancing and Third-Person Self-Talk for emotional regulation	g = 0.52, k = 230 studies	Moran, T., & Eyal, T. (2022). Emotion regulation by psychological dist	Not mapped
Progressive Muscle Relaxation for anxiety and insomnia	SMD = -1.74, 95% CI [-2.14, -1	Donato, K. O., Falcão, L., Nishizima, A., Oliveira, A. S., et al. (202	Not mapped
Biofeedback for stress and pain management	g = 0.81, k = 24 studies	Goessl, V. C., Curtiss, J. E., & Hofmann, S. G. (2017). The effect of	Not mapped
Tai Chi for balance cognition and mood in older adults	Hedge’s g = 0.90, p = 0.043	Wayne, P. M., Walsh, J. N., Taylor-Piliae, R. E., Wells, R. E., Papp,	Not mapped
Music Therapy for depression anxiety and cognitive function	SMD = -0.66, 95% CI [-0.86, -0	Tang, Q., Huang, Z., Zhou, H., & Ye, P. (2020). Effects of music thera	Not mapped
Pet-Assisted Therapy (Animal-Assisted Interventions) for mental health	g = -0.67, p < .001, k = 14 st	Sim, S. Q., Liu, Z., Wu, Z., Wang, S., & Nisa, C. (2025). Canine-assis	Not mapped
Laughter Therapy and Humor Interventions for stress and immunity	SMD = -1.41, 95% CI [-2.04, -0	Li, L., Cui, Y., Guo, X., Zhou, Y., & Tang, Y. (2026). Laughter interv	Not mapped
Bibliotherapy (guided self-help reading) for depression and anxiety	d = 0.77, k = 17 studies	Gregory, R. J., Canning, S. S., Lee, T. W., & Wise, J. C. (2004). Cogn	Not mapped

Therapy	Effect Size	Primary Citation	Target Lines
Aerobic Exercise for depression (dose-response relationship)	SMD = -0.93, 95% CI [-1.25, -0	Tian, S., Liang, Z., Qui, F., Yu, Y., Wang, C., Zhang, M., & Wang, X.	Not mapped
Resistance Training for depression and anxiety	d = 0.66, 95% CI [0.48, 0.83],	Gordon, B. R., McDowell, C. P., Hallgren, M., Meyer, J. D., Lyons, M.,	Not mapped
Mediterranean Diet for depression prevention	SMD = -0.53, 95% CI [-0.90, -0	Bizzozero-Peroni, B., Martínez-Vizcaíno, V., Fernández-Rodríguez, R.,	Not mapped
Social Prescribing and Community Referral for loneliness	moderate reductions in lonelin	Wang, Y., Xu, Y., Gao, L., & Bai, X. (2026). Impact of social prescrib	Not mapped
Polyvagal exercises and vagal toning for anxiety and emotional regulation	SMD = -0.53, 95% CI [-0.74, -0	Lin, Z., Wang, D., Nie, Y., Tan, C., & Liu, Z. (2026). Transcutaneous	Emotional Regulation, Body Awarenes
Psilocybin-assisted therapy	d = 2.5, 95% CI [1.4, 3.5], k	Davis, A. K., Barrett, F. S., May, D. G., Cosimano, M. P., Sepeda, N.	Emotional Regulation, Meaning-Makin
MDMA-assisted therapy	d = 0.70, 95% CI [-26.94, -20.	Mitchell, J. M., Ot’alora G., M., van der Kolk, B., Shannon, S., Bogen	Trauma Integration, Emotional Regul
Neurofeedback (EEG biofeedback)	SMD = 0.80, k = 10 studies	Van Doren, J., Arns, M., Heinrich, H., Vollebregt, M. A., Strehl, U.,	Attention Regulation, Emotional Reg
Float tanks and sensory deprivation (REST) for anxiety and pain	d = 0.65, k = 27 studies	van Dierendonck, D., & te Nijenhuis, J. (2005). Flotation restricted e	Emotional Regulation, Body Awarenes
Ketamine-assisted therapy for treatment-resistant depression	d = -0.73, 95% CI [-0.91, -0.5	Nikolin, S., Rodgers, A., Schwaab, A., Bahji, A., Zarate, C. A., Jr, V	Emotional Regulation, Cognitive Fle

Therapy	Effect Size	Primary Citation	Target Lines
Transcranial magnetic stimulation (TMS)	OR = 3.38, 95% CI [2.24, 5.12]	Brunoni, A. R., Chaimani, A., Moffa, A. H., et al. (2017). Repetitive	Emotional Regulation, Cognitive Fle
Eye Movement Desensitization and Reprocessing (EMDR)	g = 0.93, 95% CI [0.67, 1.18],	Cuijpers, P., Veen, S. C., Sijbrandij, M., Yoder, W., & Cristea, I. A.	Trauma Integration, Emotional Regul
Dialectical Behavior Therapy (DBT)	d = 0.324, 95% CI [-0.471, -0.	DeCou, C. R., Comtois, K. A., & Landes, S. J. (2019). Dialectical beha	Emotional Regulation, Interpersonal
Schema Therapy (ST)	g = 0.359, k = 15 studies	Zhang, K., Hu, X., Ma, L., Xie, Q., Wang, Z., & Li, X. (2023). The eff	Emotional Regulation, Interpersonal
Narrative Therapy	SMD = -1.64, 95% CI [-1.95, -1	Hu, G., Han, B., Gains, H., & Jia, Y. (2024). Effectiveness of narrati	Identity Coherence, Meaning-Making,
Solution-Focused Brief Therapy (SFBT)	g = 1.17, k = 72 studies	Vermeulen-Oskam, E., Franklin, C., Van't Hof, L. P. M., & Roest, J. J.	Volitional Control, Meaning-Making,
Compassion-Focused Therapy (CFT)	g = 0.51, 95% CI [0.33, 0.69],	Wakelin, K. E., Perman, G., & Simonds, L. M. (2022). Effectiveness of	Emotional Regulation, Identity Cohe
Interpersonal Psychotherapy (IPT)	d = 0.63, 95% CI [0.36, 0.90],	Cuijpers, P., Geraedts, A. S., van Oppen, P., Andersson, G., Markowitz	Emotional Regulation, Interpersonal
Behavioral Activation (BA)	d = -0.74, 95% CI [-0.91, -0.5	Ekers, D., Webster, L., Van Straten, A., Cuijpers, P., Richards, D., &	Emotional Regulation, Volitional Co
Mindfulness-Based Cognitive Therapy (MBCT)	HR = 0.69, 95% CI [0.58, 0.82]	Kuyken, W., Warren, F. C., Taylor, R. S., Whalley, B., Crane, C., Bond	Emotional Regulation, Cognitive Fle
Trauma-Focused Cognitive Behavioral	g = 1.14, 95% CI [0.97, 1.30]	Thielemann, J. F. B., Kasparik, B., König, J.,	Emotional Regulation,

Therapy	Effect Size	Primary Citation	Target Lines
Therapy (TF-CBT)		Unterhitzenberger, J.,	Trauma Integr
Existential Therapy and Logotherapy	d = 0.65, k = 6 studies (for p	Vos, J., Craig, M., & Cooper, M. (2015). Existential therapies: A meta	Meaning-Making, Identity Coherence,
Positive Psychology Interventions (PPIs)	g = 0.39 for wellbeing, g = -0	Carr, A., Cullen, K., Keeney, C., Canning, C., Mooney, O., Chinseallai	Meaning-Making, Emotional Regulatio
Attachment-Based Therapy (Emotionally Focused Therapy)	d = 0.93	Spengler, P. M., Lee, N. A., Wiebe, S. A., & Wittenborn, A. K. (2024).	Emotional Regulation, Interpersonal
Cognitive Behavioral Therapy for Insomnia (CBT-I)	g = 0.98, 95% CI [0.81, 1.16],	Scott, A. J., Webb, T. L., Rowse, G., & et al. (2025). Cognitive Behav	sleep quality, emotional regulation
Sleep restriction therapy for chronic insomnia	g = -0.93, 95% CI [-1.15, -0.7	Maurer, L. F., Schneider, J., Miller, C. B., Espie, C. A., & Kyle, S.	sleep quality, sleep efficiency, at
Flow state training and optimal experience interventions for peak performance	r = 0.31, 95% CI [0.24, 0.38],	Harris, D. J., Allen, K. L., Vine, S. J., & Wilson, M. R. (2021). A sy	attention regulation, volitional co
Spaced repetition and retrieval practice for long-term memory and learning	g = 0.61, k = 118 experiments	Adesope, O. O., Trevisan, D. A., & Sundararajan, N. (2017). Rethinking	memory consolidation, cognitive enh
Gratitude interventions (three good things) for subjective wellbeing	g = 0.17, 95% CI [0.14, 0.20],	Choi, H., et al. (2025). A meta-analysis of the effectiveness of grati	emotional regulation, meaning-makin
Goal-setting theory (SMART goals)	d = 0.80, k = 23 effect sizes	Kleingeld, A., van Mierlo, H., & Arends, L. (2011). The effect of goal	volitional control, motivation, att
Implementation intentions (if-then	d = 0.65, k = 94 studies	Gollwitzer, P. M., & Sheeran, P. (2006).	volitional control, cognitive flexi

Therapy	Effect Size	Primary Citation	Target Lines
planning) for behavior change		Implementation intentions and	
Habit stacking and environmental design for behavior modification	d = 0.65, 95% CI [0.55, 0.75],	Gollwitzer, P. M., & Sheeran, P. (2006). Implementation intentions and	volitional control, cognitive flexi
Couples therapy (Gottman Method) for relationship satisfaction	d = 0.65, 95% CI [0.45, 0.85],	Kernová, L., Halamová, J., & Deriglazov, D. (2025). Effectiveness of d	emotional regulation, interpersonal
Emotionally Focused Couples Therapy	d = 0.93, 95% CI [0.75, 1.12],	Spengler, P. M., Lee, N. A., Wiebe, S. A., & Wittenborn, A. K. (2024).	emotional regulation, trauma integr
Group therapy and support groups for social connection and recovery	d = -0.59, 95% CI [-0.98, -0.2	Pfeiffer, P. N., Heisler, M., Piette, J. D., Feldman, M. A., & Valenst	social connection, emotional regula
Mindfulness-Based Stress Reduction (MBSR) for chronic pain and anxiety	Non-inferiority margin = -0.49	Hoge, E. A., Bui, E., Mete, M., Dutton, M. A., Baker, A. W., & Simon,	emotional regulation, body awarenes
12-Step Programs (AA/NA)	RR = 1.21, 95% CI [1.03, 1.42]	Kelly, J. F., Humphreys, K., & Ferri, M. (2020). Alcoholics Anonymous	Addiction, Behavioral, Meaning/Purp
Contingency Management	d = 0.70, 95% CI [0.49, 0.92],	Bolívar, H. A., Klemperer, E. M., Coleman, S. R. M., DeSarno, M., Skel	Addiction, Impulsivity, Motivation,
Motivational Interviewing combined with CBT for dual diagnosis	g = 0.27, 95% CI [0.13, 0.41]	Riper, H., Andersson, G., Hunter, S. B., de Wit, J., Berking, M., & Cu	Addiction, Depression, Self-Efficac
Transcendental Meditation ™	d = -4.7 mm Hg (95% CI [-7.4,	Anderson, J. W., Liu, C., & Kryscio, R. J. (2008). Blood pressure resp	Cognitive, Emotional, Physical

Therapy	Effect Size	Primary Citation	Target Lines
Acupuncture for chronic pain depression and anxiety	SMD = -0.72, 95% CI [-0.91, -0	Zhao, H., Gao, Y., Zhang, K., Tang, C., & Shen, W. (2026). The efficac	Cognitive, Emotional, Somatic
Light therapy (phototherapy) for seasonal affective disorder and circadian rhythm	SMD = -0.37, 95% CI [-0.63, -0	Pjrek, E., Friedrich, M. E., Cambioli, L., Dold, M., Jäger, F., Komoro	Emotional Regulation, Somatic Aware
Gut microbiome interventions (probiotics) for depression and anxiety	SMD = -0.96, 95% CI [-1.31, -0	Asad, A., Kirk, M., Zhu, S., Dong, X., & Gao, M. (2025). Effects of pr	Cognitive, Emotional, Physical
Walking in nature (green exercise) for mood and attention restoration	SMD = -0.40, 95% CI [-0.56, -0	Hu, G., Luo, Q., Zhang, P., Zeng, H., & Ma, X. (2025). Effects of urba	Emotional Regulation, Attention, St
Executive coaching for leadership development and self-awareness	g = 0.59, k = 37 studies	de Haan, E., & Nilsson, V. O. (2023). What can we know about the effec	Cognitive, Emotional, Interpersonal
Occupational therapy for functional recovery and daily living skills	SMD = 0.72, 95% CI [0.40, 1.05	Hua, S., et al. (2025). Occupational therapy improves functional recov	Physical, Behavioral, Social
Mentalization-Based Treatment (MBT)	g = -1.08, 95% CI [-1.38, -0.7	Hajek Gross, C., Oehlke, S. M., Prillinger, K., & Plener, P. L. (2024)	Emotional, Interpersonal, Self-Iden
Mindful Self-Compassion (MSC)	Hedges' g = 0.51, 95% CI [0.33	Wakelin, K. E., Perman, G., & Simonds, L. M. (2022). Effectiveness of	Self-Criticism, Shame, Self-Compass
Cognitive Processing Therapy (CPT)	Hedges' g = 1.24, k = 11 studi	Asmundson, G. J. G., Thorisdottir, A. S., Roden-Foreman, J. W., Baird,	Cognitive, Emotional, Moral

Therapy	Effect Size	Primary Citation	Target Lines
Prolonged Exposure therapy	Hedges's g = 1.08, 95% CI [0.6	Powers, M. B., Halpern, J. M., Ferenschak, M. P., Gillihan, S. J., & F	Emotional, Cognitive, Behavioral
Sandplay therapy and play therapy	g = 1.10, k = 40 studies	Wiersma, J. K., Freedle, L. R., McRoberts, R., & Solberg, K. B. (2022)	Emotional expression, Social commun
Virtual Reality Exposure Therapy (VRET)	g = 0.90 (vs waitlist), g = 0.	Carl, E., Stein, A. T., Levihn-Coon, A., Pogue, J. R., Rothbaum, B., E	Cognitive, Emotional, Behavioral
Parent-Child Interaction Therapy (PCIT)	d = -0.87, 95% CI [-1.10, -0.6	Valero-Aguayo, L., Rodríguez-Bocanegra, M., & Mendo-Lázaro, S. (2021).	Behavioral, Emotional, Interpersona
Reminiscence therapy for older adults with depression and dementia	d = -0.54, 95% CI [-0.85, -0.2	Park, K., Lee, S., Yang, J., Song, T., & Hong, G. R. S. (2019). A syst	Cognitive Line, Emotional Line
Grief therapy and complicated grief treatment	SMD = -0.53, 95% CI [-0.84, -0	Wittouck, C., Van Autreve, S., De Jaegere, E., Portzky, G., & van Heer	Emotional, Cognitive, Interpersonal
Cancer survivorship interventions for post-treatment adjustment	g = 0.35, 95% CI [0.16, 0.54],	Salsman, J. M., Pustejovsky, J. E., Schueller, S. M., Hernandez, R., B	Emotional, Cognitive, Interpersonal
Trauma-Sensitive Yoga	g = 0.56, 95% CI [0.29, 0.82],	van de Kamp, M. M., Scheffers, M., Hatzmann, J., Emck, C., Cuijpers, P	Somatic, Emotional, Trauma/Shadow
Horticultural therapy (therapeutic gardening)	SMD = -0.86, 95% CI [-1.22, -0	Xu, M., Lu, S., Liu, J., & Xu, F. (2023). Effectiveness of horticultur	Emotional regulation, social connec
Acceptance-based approaches for chronic	d = 0.50 (medium effect	Ye, L., Li, Y., Deng, Q., Zhao, X., Zhong, L., &	Cognitive, Emotional,

Therapy	Effect Size	Primary Citation	Target Lines
illness and disability adjustment	size)	Yang, L. (2024). Acce	Physical

Moderate (57 therapies)

Therapy	Effect Size	Primary Citation	Target Lines
Acceptance and Commitment Therapy (ACT)	g = 0.22, 95% CI [0.01, 0.23],	Fang, S., & Ding, D. (2023). The differences between acceptance and co	Not mapped
Mindfulness-Based Stress Reduction (MBSR)	g = 0.53, 95% CI [0.41, 0.64],	Khoury, B., Sharma, M., Rush, S. E., & Fournier, C. (2015). Mindfulnes	Not mapped
Gratitude Practice	g = 0.19, 95% CI [0.15, 0.22],	Choi, H., Cha, Y., McCullough, M. E., Coles, N. A., & Oishi, S. (2025)	Not mapped
Expressive Writing and Journaling (Pennebaker paradigm)	r = .075, k = 146 studies	Frattaroli, J. (2006). Experimental disclosure and its moderators: a m	Not mapped
Motivational Interviewing for behavior change	g = 0.28, k = 119 studies	Lundahl, B. W., Kunz, C., Brownell, C., Tollefson, D., & Burke, B. L.	Not mapped
Affect Labeling (Name It to Tame It) for emotional regulation	g = -0.43, 95% CI [-0.68, -0.1	Rizk, M. A. (2024). Systematic review and meta-analysis of affect labe	Not mapped
Mastery Experiences in Off-Job Time for recovery and wellbeing	Positive meta-analytic associa	Headrick, L., Newman, D. A., Park, Y. A., & Liang, Y. (2023). Recovery	Not mapped
Psychological Detachment From Work for burnout prevention	d = 0.36, k = 30 studies, 34 i	Karabinski, T., Haun, V. C., Nübold, A., Wendsche, J., & Wegge, J. (20	Not mapped
Gestalt Empty Chair technique for unfinished business and grief	d = 0.40	Pascual-Leone, A., & Baher, T. (2023).	Not mapped

Therapy	Effect Size	Primary Citation	Target Lines
		Chairwork in individual psychot	
Qigong for chronic pain and mental health	SMD = -0.27, 95% CI [-0.52, -0	Zou, L., Yeung, A., Li, C., Wei, G. X., Chen, K. W., Kinser, P. A., ..	Not mapped
Art Therapy for trauma and emotional expression	g = 0.89, p = 0.052, k = 21 st	Maddox, G. A., Bodner, G. E., Christian, M. W., & Williamson, P. (2024	Not mapped
Cold Water Immersion and Cold Exposure for mood and resilience	g = -0.42, 95% CI [-0.99, 0.15	Ngandeu Schepanski, S., Batta, F., Schröter, M., Seifert, G., & Michal	Not mapped
Sauna and Heat Therapy for cardiovascular and mental health	MD = -3.94 mmHg, 95% CI [-7.22	Pizzey, F. K., Smith, E. C., & Ruediger, S. L. (2021). The effect of h	Not mapped
Volunteering and Prosocial Behavior for wellbeing and longevity	r = .13 (Cohen’s d = 0.26), K	Hui, B. P. H., Ng, J. C. K., Berzaghi, E., Cunningham-Amos, L. A., & K	Not mapped
Sleep Hygiene Education for insomnia prevention	MD = 3.4, 95% CI [2.08, 4.64],	Ruan, J. Y., Liu, Q., Chung, K. F., Ho, K. Y., & Yeung, W. F. (2025).	Not mapped
Loving-Kindness Meditation (Metta) for positive affect and social connection	g = 0.44, 95% CI [0.21, 0.67],	Petrovic, J., Mettler, J., Cho, S., & Heath, N. L. (2024). The effects	Not mapped
Somatic Experiencing (SE)	d = 0.94 to 1.26, n = 63 parti	Brom, D., Stokar, Y., Lawi, C., Nuriel-Porat, V., Ziv, Y., Lerner, K.,	Trauma Integration, Body Awareness,
Internal Family Systems (IFS) therapy	d = -4.46 (CAPS) and d = -3.05	Hodgdon, H. B., Anderson, F. G., Southwell, E., Hrubec, W., & Schwartz	Trauma Integration, Emotional Regul

Therapy	Effect Size	Primary Citation	Target Lines
Cold exposure and cold water immersion (Wim Hof method)	r = 0.25, 95% CI [-7.197, -1.3	Blades, R., Mendes, W. B., Don, B. P., Mayer, S. E., Dileo, R., O'Brya	Emotional Regulation, Stress Resili
Holotropic Breathwork	g = -0.35, 95% CI [-0.55, -0.1	Fincham, G. W., Strauss, C., Montero-Marin, J., & Cavanagh, K. (2023).	Emotional Regulation, Trauma Integr
Forest bathing (Shinrin-yoku)	MD = -0.08 µg/dl [95% CI -0.11	Antonelli, M., Barbieri, G., & Donelli, D. (2019). Effects of forest b	Emotional Regulation, Body Awarenes
Motivational Enhancement Therapy (MET)	g = 0.28, k = 119 studies	Lundahl, B. W., Kunz, C., Brownell, C., Tollefson, D., & Burke, B. L.	Volitional Control, Cognitive Flexi
Acceptance-Based Behavioral Therapy (ABBT)	d = 1.27 to 1.61 (pre to post	Hayes-Skelton, S. A., Roemer, L., & Orsillo, S. M. (2013). A randomize	Emotional Regulation, Cognitive Fle
Lucid dreaming training for nightmare reduction and creativity	d = 0.48, 95% CI [0.36, 0.60],	Augedal, A. W., Hansen, K. S., Kronhaug, C. R., Harvey, A. G., & Palle	emotional regulation, trauma integr
Deliberate practice and expertise development for skill acquisition	r = 0.35 (12% of variance), k	Macnamara, B. N., Hambrick, D. Z., & Oswald, F. L. (2014). Deliberate	cognitive flexibility, attention re
Meditation and contemplative practices for attention and executive function	g = 0.203, 95% CI [0.143, 0.26	Sumantry, D., & Stewart, K. E. (2021). Meditation, mindfulness, and at	attention regulation, executive fun
Journaling and expressive writing for emotional processing and self-awareness	d = 0.47, k = 13 studies	Smyth, J. M. (1998). Written emotional expression: effect sizes, outco	emotional regulation, trauma integr

Therapy	Effect Size	Primary Citation	Target Lines
Wilderness therapy and adventure-based counseling for adolescents and adults	g = 0.47, k = 197 studies	Bowen, D. J., & Neill, J. T. (2013). A meta-analysis of adventure ther	emotional regulation, interpersonal
Cognitive Remediation Therapy	d = 0.45, k = 40 studies	Wykes, T., Huddy, V., Cellard, C., McGurk, S. R., & Czobor, P. (2011).	cognitive flexibility, attention re
Mindfulness-Based Relapse Prevention	OR = 0.72, 95% CI [0.46, 1.13]	Grant, S., Colaiaco, B., Motala, A., Shanman, R., Booth, M., Sorbero,	Cognitive, Emotional, Behavioral
Yoga (Hatha/Vinyasa) for depression, anxiety, and PTSD	SMD = -0.51, 95% CI [-0.68, -0	Nejadghaderi, S. A., et al. (2024). Efficacy of yoga for posttraumatic	Emotional Regulation, Somatic Aware
Breathwork (pranayama and controlled breathing) for anxiety and autonomic regulation	g = -0.32, 95% CI [-0.49, -0.1	Fincham, G. W., Strauss, C., Montero-Marin, J., & Cavanagh, K. (2023).	Emotional, Physical, Spiritual
Body scan meditation for chronic pain and body awareness	Hedge’s g = 0.268, 95% CI [0.0	Gan, R., Zhang, Y., & Chen, S. (2022). The effects of body scan medita	Physical, Emotional
Feldenkrais Method	d = -1.14, 95% CI [-1.78, -0.4	Berland, R., et al. (2022). Effects of the Feldenkrais Method as a Phy	Physical, Emotional
Massage therapy for anxiety depression and cortisol reduction	d = 0.73 for trait anxiety, d	Moyer, C. A., Rounds, J., & Hannum, J. W. (2004). A Meta-Analysis of M	Emotional, Physical
Omega-3 fatty acid supplementation for depression and cognitive function	SMD = -0.28, 95% CI [-0.46, -0	Liao, Y., Xie, B., Zhang, H., He, Q., Guo, L., Subramanieapillai, M.,	Cognitive, Emotional

Therapy	Effect Size	Primary Citation	Target Lines
Vitamin D supplementation for depression and immune function	d = 0.58, 95% CI [0.45, 0.72],	Vellekkatt, F., & Menon, V. (2019). Efficacy of vitamin D supplementat	Cognitive, Emotional, Physical
Time-Restricted Eating	WMD = -1.60 kg, 95% CI [-2.27,	Liu, L., et al. (2022). Metabolic Efficacy of Time-Restricted Eating i	Physical Health, Cognitive Function
High-Intensity Interval Training (HIIT)	SMD = -0.40, 95% CI [-0.60, -0	Tao, Y., et al. (2024). Effects of high-intensity interval training on	Cognitive, Emotional, Physical
Cognitive Behavioral Coaching	g = 0.499, k = 26 studies	Tomoiagă, C., & David, O. (2023). Is cognitive-behavioral coaching an	Cognitive, Emotional
Career counseling and vocational guidance for purpose and direction	d = 0.352, k = 57 studies	Whiston, S. C., Li, Y., Mitts, N. G., & Wright, L. (2017). Effectivene	Cognitive, Emotional, Interpersonal
Psychoeducation for mental health literacy and self-management	SMD = 0.42, 95% CI [-0.10, 0.7	Yeo, G. H., et al. (2024). The Effect of Digital Mental Health Literac	Cognitive, Emotional, Behavioral
Radically Open Dialectical Behavior Therapy (RO-DBT)	d = 1.03, k = 1 study (RefraME	Lynch, T. R., Hempel, R. J., Whalley, B., Byford, S., Chamba, R., Clar	Cognitive Inflexibility, Emotional
Acceptance and Commitment Training (ACT)	g = 0.301, 95% CI [0.122, 0.48	Prudenzi, A., Graham, C. D., Clancy, F., Hill, D., O'Driscoll, R., Day	Psychological flexibility, Cognitiv
Stellate Ganglion Block	Adjusted mean CAPS-5 score cha	Rae Olmsted, K. L., Bartoszek, M., Mulvaney, S., McLean, B., Turabi, A	Physiological hyperarousal, psychol
Creative arts therapies (drama therapy and psychodrama) for trauma	SMD = -1.98, 95% CI [-3.8, -0.	Wang, J., Zhang, B., Yahaya, R., & Abdullah, A. B. (2025). Colors of t	Emotional, Interpersonal, Cognitive
Binaural beats and auditory stimulation for	g = 0.45, k = 22 studies	Garcia-Argibay, M., Santed, M. A., & Reales,	Cognitive, Emotional

Therapy	Effect Size	Primary Citation	Target Lines
focus and relaxation		J. M. (2019). Efficacy of	
Gamification and serious games for cognitive training and motivation	Hedges' g = 0.72, p = .002, k	Vermeir, J. F., White, M. J., Johnson, D., Crombez, G., & Van Ryckeghe	Cognitive, Emotional, Motivational
Digital CBT and app-based mental health interventions	g = 0.28, p < 0.001, NNT = 11.	Linardon, J., Torous, J., Firth, J., Cuijpers, P., Messer, M., & Fulle	Cognitive, Emotional, Behavioral
Peer support and lived experience mentoring for recovery	SMD = 0.29, 95% CI [0.13, 0.45	Jambawo, S. M., Owolewa, R., & Jambawo, T. T. (2024). The effectivenes	LR (Lower Right) - Social/Interpers
Emotion coaching (Gottman parenting) for child emotional development	d = 0.31, 95% CrI [0.20, 0.41]	Zahl-Olsen, R., Severinsen, L., Stiegler, J. R., Fernee, C. R., Simhan	Emotional Regulation, Parent-Child
Meaning-Centered Psychotherapy	d = -0.96 for meaning in life,	Kang, K. A., Han, S. J., Lim, Y. S., & Kim, S. J. (2019). Meaning-Cent	Meaning, Purpose, Spiritual Well-be
Cardiac rehabilitation programs for recovery and psychological wellbeing	d = 0.297 (mental health treat	Rutledge, T., Redwine, L. S., Linke, S. E., & Mills, P. J. (2013). A m	Emotional, Physical, Social
Autism spectrum interventions (social skills training) for adults	g = 0.93, 95% CI [0.55, 1.30],	Dubreucq, J., Haesebaert, F., Plasse, J., Dubreucq, M., & Franck, N. (	Social, Emotional, Interpersonal
Dyslexia interventions and reading remediation for adults	d = 0.38, k = 72 studies	Toffalini, E., Giofrè, D., Pastore, M., Carretti, B., Friso, G., & Pal	Cognitive Line, Educational Line
Equine-assisted therapy for trauma PTSD and emotional regulation	d = 12.46 (mean difference), 9	Provan, M., & Provan, C. (2024). Are equine-assisted services benefici	Emotional Regulation, Trauma Recove

Therapy	Effect Size	Primary Citation	Target Lines
Dignity Therapy	SMD = -0.38, 95% CI [-0.56, -0	Lee, J. L., & Jeong, Y. (2023). The Effect of Dignity Therapy on Termi	Meaning & Purpose, Emotional Well-b

Emerging (5 therapies)

Therapy	Effect Size	Primary Citation	Target Lines
Coherence Therapy (Memory Reconsolidation)	d = 0.23 on procrastination (w	Rice, K. G., Neimeyer, G. J., & Taylor, J. M. (2011). Efficacy of Cohe	Emotional Regulation, Trauma Integr
Alexander Technique	SMD = -0.34, 95% CI [-0.87, 0.	Qin, D., et al. (2024). Effects of the Alexander technique on pain and	Physical Health, Emotional Regulati
Psychedelic Integration Therapy	N/A	Bathje, G. J., Majeski, E., & Kudowor, M. (2022). Psychedelic integrat	Meaning-making, Emotional Processin
Drum circles and rhythmic entrainment for social bonding and mood	d = -2.14 to -3.41 (depression	Fancourt, D., Perkins, R., Ascenso, S., Carvalho, L. A., Steptoe, A.,	Social, Emotional, Interpersonal
Philosophical counseling and Socratic dialogue for existential questions	d = -1.51 (b* = -1.51, p = .01	Braun, J. D., Strunk, D. R., Sasso, K. E., & Cooper, A. A. (2015). The	Meaning/Purpose, Cognitive/Beliefs,

Other (3 therapies)

Therapy	Effect Size	Primary Citation	Target Lines
Dual N-Back training for working memory and fluid intelligence	g = 0.16, 95% CI [0.06, 0.25],	Soveri, A., Antfolk, J., Karlsson, L., Salo, B., & Laine, M. (2017). W	cognitive flexibility, attention re
Neurolinguistic Programming (NLP) for behavior change and communication	N/A (No significant effect fou	Sturt, J., Ali, S., Robertson, W., Metcalfe, D., Grove, A., Bourne, C.	communication skills, emotional reg
Harm Reduction approaches for addiction and risky behavior	d = 0.03, 95% CI [-0.08, 0.14]	O'Leary, C., Ralphs, R., Stevenson, J., Smith, A., Harrison, J., Kiss,	Addiction, Risky Behavior

2. Therapy-to-Line Mapping

Coverage: 21 of 32 Lines

Line	# Therapies	Top Interventions
Intrapersonal	13	Self-Compassion (Neff), Psychodynamic Therapy, MDMA-Assisted Therapy
Meta-Cognitive	11	Clinical Hypnosis, Cognitive Defusion (ACT), Self-Distancing Third-Person Self-Talk (Kross)
Interoceptive	9	Progressive Muscle Relaxation (PMR), Biofeedback HRV, Polyvagal Exercises
Volitional	8	Aerobic Exercise, Resistance Training, Neurofeedback
Empathic	4	Forgiveness Interventions (Wade), Music Therapy, Pet-Assisted Therapy
Resilient	4	Laughter Therapy, Positive Psychology Interventions (Strengths-Based), Prolonged Exposure (PE)
Adaptive	4	Bibliotherapy (Guided Self-Help), EMDR, Psilocybin-Assisted Therapy
Interpersonal	4	Social Prescribing, Interpersonal Therapy (IPT), Gottman Method Couples Therapy
Reflective	2	Narrative Therapy, Gratitude Interventions
Strategic	2	Solution-Focused Brief Therapy (SFBT), Contingency Management
Existential	2	Logotherapy (Frankl), Meaning-Oriented Grief Therapy (Neimeyer)
Tactical	2	Goal-Setting Theory (SMART goals), Light Therapy
Kinesthetic	1	Dance Therapy (Wu 2021)
Logical	1	Mediterranean Diet
Integrative	1	Ketamine-Assisted Therapy

Line	# Therapies	Top Interventions
Systematic	1	Habit Stacking
Linguistic	1	Cognitive Stimulation Therapy (CST)
Parenting	1	Parent-Child Interaction Therapy (PCIT)
Naturalistic	1	Forest Bathing (Shinrin-yoku)

11 Gap Lines (No Therapy Coverage)

Line	Status	Recommended Intervention Type
Mathematical	GAP	Quantitative reasoning training, statistical thinking curricula
Spatial	FILLED (Uttal 2013, g=0.47)	Mental rotation training, video game spatial training
Philosophical	GAP	Socratic dialogue, ethics coursework, philosophical counseling
Intuitive	GAP	Recognition-Primed Decision training, perceptual learning
Musical	FILLED (Olszewska 2021)	Adult instrumental instruction, rhythmic training
Architectural	GAP	Systems design training, complex project planning
Adversarial	GAP	Negotiation training, red-team thinking, game theory
Aesthetic	GAP	Art appreciation curricula, design thinking training
Influence	GAP	Public speaking, leadership communication, rhetoric
Seduction	GAP	Social calibration training, courtship skill development
Financial-Self-Mgmt	FILLED (Kaiser 2016, g=0.27)	Financial literacy programs, behavioral finance training

3. Intelligence Line Validity — All 32 Lines

Perplexity-Verified Findings (August 6, 2026)

Line	Construct Validity	Best Instrument	Key DOI	Independence from g
Philosophical	VALIDATED	Reflective Judgment Interview (King & Kitchener 1981)	10.1207/s15326985ep3901_2	Predicts beyond verbal ability/IQ
Tactical	VALIDATED	Action Control Scale ACS-90 (Kuhl 1994)	10.1037/0021-9010.85.2.250	Distinct from cognitive ability AND personality
Architectural	VALIDATED	MicroDYN (Greiff et al. 2012)	10.3389/fpsyg.2015.01669	5% incremental variance beyond Gf
Parenting	VALIDATED	PRFQ (Luyten et al. 2017)	10.1371/journal.pone.0176218	Predicts child attachment; alpha 0.70-0.82
Seduction	PARTIALLY VALIDATED	Mating Intelligence Scale (Geher & Kaufman 2007)	10.1177/2158244019867857	CFI 0.94-0.96; predicts sexual behavior
Community-Founding	VALIDATED (with caveat)	SCI-2 (Chavis et al. 2008)	10.1037/t33090-000	Alpha 0.94; but weaker in replication (0.50-0.69)

Full 32-Line Audit Results

Line	Construct Validity	Distinctness	Measurement Support
Logical intelligence line	WELL-ESTABLISHED	OVERLAPPING (with mathematical, analytic	WAIS-IV (Matrix Reasoning, Figure Weights), Raven's Progress
Mathematical intelligence line	WELL-ESTABLISHED	OVERLAPPING (with Logical intelligence l	Woodcock-Johnson IV (Quantitative Knowledge/Gq), WAIS-IV (Ar
Spatial intelligence line	WELL-ESTABLISHED	DISTINCT	Thurstone's Primary Mental Abilities (Spatial factor), Shepa
Linguistic intelligence line	WELL-ESTABLISHED	DISTINCT	WAIS Verbal Comprehension Index (VCI), CHC model (Crystalliz
Volitional intelligence line	WELL-ESTABLISHED	OVERLAPPING (with conscientiousness and	Brief Self-Control Scale (Tangney et al.), Grit Scale (Duckw
Meta-Cognitive intelligence line	WELL-ESTABLISHED	MOSTLY DISTINCT	Metacognitive Awareness Inventory (MAI), State Metacognitive
Intrapersonal intelligence line	SUPPORTED	OVERLAPPING (with Meta-Cognitive and Ref	Gardner's Multiple Intelligences theory; Emotional Intellige
Reflective intelligence line	PARTIALLY SUPPORTED	OVERLAPPING (with Intrapersonal and Meta	Schön's reflective practice research, Kember's Reflective Th
Existential intelligence line	PARTIALLY SUPPORTED	DISTINCT	Existential Intelligence Scale (EIS), Meaning in Life Questi

Line	Construct Validity	Distinctness	Measurement Support
Philosophical intelligence line	WEAK SUPPORT	OVERLAPPING (with Logical and Existentia	Defining Issues Test (moral reasoning), Watson-Glaser Critic
Integrative intelligence line	SUPPORTED	OVERLAPPING (with cognitive flexibility	Integrative Complexity Coding Manual (Suedfeld & Tetlock), S
Interpersonal intelligence line	WELL-ESTABLISHED	DISTINCT	MSCEIT (Mayer-Salovey-Caruso Emotional Intelligence Test), B
Empathic intelligence line	WELL-ESTABLISHED	DISTINCT	Interpersonal Reactivity Index (IRI), Empathy Quotient (EQ),
Intuitive intelligence line	PARTIALLY SUPPORTED	OVERLAPPING (with cognitive, emotional,	Rational-Experiential Inventory (REI), Preference for Intuit
Musical intelligence line	WELL-ESTABLISHED	DISTINCT	Gardner’s Multiple Intelligences framework, Advanced Measure
Kinesthetic intelligence line	WELL-ESTABLISHED	DISTINCT	Bruininks-Oseretsky Test of Motor Proficiency (BOT-2), Movem
Naturalistic intelligence line	PARTIALLY SUPPORTED	MOSTLY DISTINCT	Gardner’s Multiple Intelligences research program; Multiple
Strategic intelligence line	PARTIALLY SUPPORTED	OVERLAPPING (with executive function, lo	Strategic Thinking Questionnaire (STQ), Maccoby’s Strategic
Tactical intelligence line	WEAK SUPPORT	OVERLAPPING (with Strategic and Volition	Sternberg’s Triarchic Theory (Practical Intelligence), Execu
Adaptive intelligence line	SUPPORTED	OVERLAPPING (with Resilient, Cognitive F	Bar-On EQ-i (Adaptability scale),

Line	Construct Validity	Distinctness	Measurement Support
			Cognitive Flexibility Inve
Resilient intelligence line	WELL-ESTABLISHED	OVERLAPPING (with Adaptive intelligence	Connor-Davidson Resilience Scale (CD-RISC), Brief Resilience
Systematic intelligence line	PARTIALLY SUPPORTED	OVERLAPPING (with logical-mathematical i	Systemizing Quotient (SQ) by Baron-Cohen, Big Five personali
Architectural intelligence line	WEAK SUPPORT	OVERLAPPING (with Spatial and Systematic	WAIS Block Design, Mental Rotation Test, and Bennett Mechani
Adversarial intelligence line	PARTIALLY SUPPORTED	OVERLAPPING (with Social Intelligence, I	MACH-IV scale, behavioral game theory paradigms, and Theory
Interoceptive intelligence line	WELL-ESTABLISHED	MOSTLY DISTINCT	Heartbeat detection tasks (Schandry task), Multidimensional
Aesthetic intelligence line	PARTIALLY SUPPORTED	OVERLAPPING (with Openness to Experience	Visual Aesthetic Sensitivity Test (VAST), Engagement with Be
Influence intelligence line	PARTIALLY SUPPORTED	OVERLAPPING (with Interpersonal and Soci	Multifactor Leadership Questionnaire (MLQ), Social Skills In
Humor intelligence line	PARTIALLY SUPPORTED	OVERLAPPING (with verbal intelligence, s	Humor Styles Questionnaire (Martin et al., 2003), Multidimen
Parenting intelligence line	WEAK SUPPORT	OVERLAPPING (with Interpersonal and Emot	Baumrind’s parenting styles research, Parenting Stress Index
Seduction intelligence line	WEAK SUPPORT	OVERLAPPING (with Social, Emotional, and	Mating Intelligence Scale (Geher & Kaufman, 2007) and evolut

Line	Construct Validity	Distinctness	Measurement Support
Community-Founding intelligence line	WEAK SUPPORT	OVERLAPPING (with Interpersonal Intellig	Multifactor Leadership Questionnaire (MLQ), Social Capital a
Financial-Self-Management intelligence line	PARTIALLY SUPPORTED	OVERLAPPING (with Logical-Mathematical,	Lusardi & Mitchell's Financial Literacy Questions (Big Three

Empirical Support for 6 "Weak" Lines (Manus Deep Research)

Philosophical intelligence

- Best instrument: Reflective Judgment Interview (RJI) King & Kitchener 1981
- Strongest research: King & Kitchener, Reflective Judgment Model (epistemic cognition tradition), finding that individuals progress through stages of reasoning about ill-structured problems where truth is uncertain.
- Psychometric data: Inter-rater reliability for RJI is high; DIT shows Cronbach alpha around .76 and longitudinal effect sizes of .80
- Defensibility: PARTIALLY (with caveats)
- Recommended framing: Frame this as "epistemic cognition" or "reflective judgment"—the measurable capacity to reason through ill-structured problems and navigate uncertainty. It develops sequentially through adulthood and is distinct from general intelligence.

TACTICAL intelligence

- Best instrument: Action Control Scale (ACS-90) - Julius Kuhl (1994)
- Strongest research: Julius Kuhl (Action Control Theory) and Michael Frese & Dieter Zapf (Action Regulation Theory). Key findings show that action-oriented individuals are better at initiating tasks, overcoming hesitation, and maintaining goal pursuit in the face of obstacles compared to state-oriented individuals.
- Psychometric data: Cronbach's alpha typically ranges from .70 to .85 across subscales (Preoccupation/Disengagement, Hesitation/Initiative, Volatility/Persistence). Construct validity supported by confirmatory factor analysis and predictive validity in goal-striving and execution tasks.
- Defensibility: YES
- Recommended framing: Frame Tactical intelligence as 'Action-State Orientation' or 'Action Regulation'—the volitional capacity to initiate, shield, and execute goal-directed actions efficiently. Ground it in Kuhl's Action Control Theory and Frese's Action Regulation Theory,

distinguishing it from strategic planning by focusing on the immediate cognitive regulation required to overcome hesitation and execute tasks.

ARCHITECTURAL intelligence

- Best instrument: MicroDYN approach (Greiff, Wüstenberg, & Funke, 2012) and Systemizing Quotient-Revised (SQ-R) (Baron-Cohen et al., 2003; Wheelwright et al., 2006)
- Strongest research: Greiff, Wüstenberg, & Funke (Complex Problem Solving / MicroDYN) and Baron-Cohen (Empathizing-Systemizing theory). Key findings show CPS is a distinct construct from working memory and intelligence, and systemizing is a distinct cognitive style measurable independently of general intelligence.
- Psychometric data: MicroDYN shows excellent reliability estimates for its dimensions and predictive validity for school grades beyond working memory capacity. SQ-R shows high internal consistency (Cronbach's alpha = 0.903).
- Defensibility: PARTIALLY (with caveats)
- Recommended framing: Frame Architectural intelligence as the capacity for Complex Problem Solving (CPS) and Systemizing. It represents the ability to explore, understand, and control complex dynamic systems (as measured by MicroDYN) and the drive to analyze and construct rule-based systems (as measured by the Systemizing Quotient). This grounds the capacity in established, validated psychometric constructs distinct from general intelligence.

PARENTING intelligence

- Best instrument: Parental Reflective Functioning Questionnaire (PRFQ) Luyten 2017
- Strongest research: Luyten, Mayes, Nijssens, & Fonagy (2017). Rooted in attachment theory and mentalization-based research, this tradition demonstrates that parental reflective functioning (the capacity to treat the infant as a psychological agent) is a distinct capacity that predicts child attachment security and socioemotional development.
- Psychometric data: Internal consistency typically ranges from acceptable to good (e.g., Cronbach's alpha .70-.85 across subscales). Test-retest reliability for related measures like PSOC is high (r = .81 over 2 months).
- Defensibility: PARTIALLY (with caveats)
- Recommended framing: Frame this not as a general "parenting" ability, but as "Parental Reflective Functioning" (PRF) or "Parental Mentalizing." This refers to the specific cognitive-emotional capacity to treat a child as a psychological agent, understand their mental states, and regulate one's own responses accordingly. This grounds the construct in attachment theory and developmental psychology with established measurement tools like the PRFQ.

SEDUCTION intelligence

- Best instrument: Mating Intelligence Scale (Geher & Kaufman, 2007) or Flirting Styles Inventory (Hall et al., 2010)
- Strongest research: Geher & Kaufman's Mating Intelligence research program (evolutionary psychology) and Jeffrey Hall's Flirting Styles research (communication studies), which demonstrate distinct, measurable individual differences in courtship behavior and romantic competence.
- Psychometric data: Flirting Styles Inventory shows high reliability (Cronbach's $\alpha > 0.80$). Mating Intelligence Scale shows high omega coefficient and high-reliability coefficients for the total scale and subscales in some validations (e.g., Najarpourian & Samavi, 2019).
- Defensibility: PARTIALLY (with caveats)
- Recommended framing: Frame this as "Mating Intelligence" or "Romantic Competence" rather than "Seduction." Ground it in evolutionary psychology (Miller's sexual selection theory) and measurable communication styles (Hall's Flirting Styles Inventory), emphasizing the cognitive and social skills required for successful courtship and mate selection.

COMMUNITY-FOUNDING intelligence

- Best instrument: Sense of Community Index 2 (SCI-2) by Chavis, Lee, & Acosta (2008)
- Strongest research: McMillan & Chavis (1986) theory of Sense of Community, and subsequent research by Chavis et al. (2008) on the SCI-2, which identifies four elements: membership, influence, meeting needs, and shared emotional connection.
- Psychometric data: Overall scale reliability (coefficient alpha) = .94; Subscale reliabilities range from .79 to .86
- Defensibility: YES (defensible)
- Recommended framing: Frame this intelligence as the capacity to foster a "Sense of Community," which is a well-established construct in community psychology. It involves the ability to cultivate membership, influence, integration and fulfillment of needs, and shared emotional connection within a group.

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4. Perplexity-Verified Findings (Prompt Set Results)

Therapy Ratings Challenged and Confirmed

Therapy	Previous Rating	Perplexity Finding	Correct Rating
Gratitude Practice	Moderate	g = 0.19 (small); some formats reach 0.40-0.60	Moderate (confirmed, but flag as small effect)
Expressive Writing	Moderate	d = 0.075-0.16 (trivial); no durable depression benefit	Should be DOWNGRADED or flagged
Nature 120-min	Moderate	Observational only; no RCT validates threshold	Reframe as “associational, not causal”

Gap Lines Filled by Perplexity

Line	Intervention Found	Effect Size	DOI
Spatial	Mental rotation training (Uttal et al. 2013 meta-analysis, 217 studies)	g = 0.47 training; g = 0.48 transfer; durable	10.1037/a0028446
Musical	Adult instrumental training (Olszewska et al. 2021)	Structural gray-matter reorganization	10.3389/fnins.2021.630829
Musical (RCT)	Train the Brain with Music, elderly adults	Audio-motor brain plasticity in 5 weeks	10.1186/S12877-020-01761-Y
Financial	Financial education (Kaiser & Menkhoff 2016, 115 studies)	g = 0.27 literacy; g = 0.09 behavior	Kaiser et al. meta-analysis
Financial	Lusardi & Mitchell (2023) review	Modest but real gains; strongest when tailored	10.1257/jep.37.4.137

Still Missing (Perplexity Pass 1 Could Not Fill)

Line	What's Needed	Status
Mathematical	Adult quantitative reasoning RCTs	UNFILLED — needs targeted search
Intuitive	Recognition-Primed Decision training RCTs	UNFILLED — needs targeted search
Aesthetic	Validated aesthetic sensitivity training	UNFILLED — needs targeted search
Adversarial	Negotiation/red-team training RCTs with effect sizes	UNFILLED — needs targeted search
Influence	Cialdini-based persuasion training RCTs	UNFILLED — needs targeted search
Matching Algorithm	Complementarity vs. similarity research	UNFILLED — needs targeted search

5. Keystone Practices — 29 Practices Audited

Rating Corrections Required

Practice	Current	Should Be	Evidence
Behavioral Activation	Emerging	**STRONG**	d = 0.87, equivalent to CBT, first-line treatment
Mindfulness/meditation	Strong	**Moderate**	Recent meta-analyses show small-to-moderate effects with replication concerns
Journaling/reflective writing	Strong	**Moderate**	Meta-analyses show small-to-moderate effects only
Psychedelic-assisted therapy	Moderate	**Strong**	Multiple RCTs, d = -1.20 for psilocybin
Nonverbal-decoding practice	Emerging	**Moderate**	Meta-analytic support (Blanch-Hartigan)
Communication & conflict skills	Moderate	**Strong**	Multiple meta-analyses (Hawkins et al.)
Parenting-skill practice	Moderate	**Strong**	Extensive evidence base
Implementation intentions	Moderate	**Strong**	Gollwitzer meta-analysis d = 0.65
Commitment devices	Strong	**Moderate**	SEED trial is real but limited generalizability
Spaced repetition	Moderate	**Strong**	Among most robust findings in cognitive psychology
Protein + fiber baseline	Strong	**Moderate**	Observational data only for mortality claims
Anchor a sense of purpose	Strong	**Moderate**	Observational cohort data, not RCTs
WOOP/mental contrasting	Moderate	**Strong**	Multiple meta-analyses (Oettingen)

Line Assignment Corrections (19 practices need remapping)

Practice	Remove (not AQAL)	Add (correct AQAL lines)
Sleep protection	memory, emotion, decision-making, self-regulation	Volitional, Intrapersonal, Strategic, Meta-Cognitive
Aerobic exercise	mood	Volitional
Mindfulness/meditation	emotion	Interoceptive, Resilient
Nature exposure	mood, attention	Intrapersonal, Volitional
Thermal stress (sauna)	health	Adaptive, Interoceptive
Psychedelic-assisted therapy	openness	Reflective, Integrative
Deliberate practice	mastery	Meta-Cognitive
Spaced repetition	memory	Meta-Cognitive
Deep-work attention blocks	attention	(keep Strategic, Volitional)
Behavioral activation	mood, self-regulation	Intrapersonal, Resilient
Cognitive restructuring	emotion	Resilient
Resistance training	mood	Volitional
Protein + fiber	health, energy	Interoceptive, Resilient
Gratitude & prosocial	mood	Interpersonal
Expressive writing	emotion	Resilient
Morning light	mood, sleep, energy	Resilient, Adaptive, Interoceptive

6. Stage Framework — 9 Stages Verified

Stage	Name	Action Logic	Claimed %	Actual %	Source	Empirically Validated
2	Impulsive	Impulsive	0.5%	**0%** (adults)	Cook-Greuter 2004	PARTIALLY (small samples)
2/3	Self-protective	Opportunist	8%	**5%**	Rooke & Torbert 2005	YES (SCTi/MAP)
3	Conformist	Diplomat	12%	**12%**	Rooke & Torbert 2005	YES (SCTi/MAP)
3/4	Self-conscious	Expert	30%	**38%** (ERROR)	Rooke & Torbert 2005	YES (SCTi/MAP)
4	Conscientious	Achiever	30%	**30%**	Rooke & Torbert 2005	YES (SCTi/MAP)
4/5	Individualist	Individualist	10%	**10%**	Rooke & Torbert 2005	YES (SCTi/MAP)
5	Autonomous	Strategist	4%	**4-4.9%**	Cook-Greuter 2004	YES (SCTi/MAP)
5/6	Construct-aware	Alchemist	<2%	**1%**	Rooke & Torbert 2005	PARTIALLY (small samples)
6	Unitive	Ironist	<1%	**<1%**	Cook-Greuter 1999	YES (SCTi/MAP, rare)

7. Matching Algorithm — 8 Components Audited

Component	Verdict	Key Finding
Complementary Mode ($^{65}/_{35}$ weights)	MOSTLY SOUND	Reasonable but weights are arbitrary; no empirical basis for $^{65}/_{35}$ specifically
Resonance Mode ($^{60}/_{40}$ weights)	MOSTLY SOUND	Homophily research supports similarity matching
Thresholds ($^{1}/_{3}$ split, 0.82 bar)	MOSTLY SOUND	0.82 is high; consider 0.70 for broader matching
Generational Adjustment (+/-8 pts)	MOSTLY SOUND	Cross-gen mentoring has support; magnitude is reasonable
Altitude Lines (5 lines only)	QUESTIONABLE	Arbitrary selection; Meta-Cognitive and Volitional should be included
Honest Framing	SOUND	Correctly states no validated matching model exists
Bottleneck Roles (Liebig/O-Ring/TOC)	MOSTLY SOUND	Conceptually valid but needs empirical grounding in human performance
Axis Modes (independence claims)	QUESTIONABLE	Humor and Adversarial g-independence claims need revision

8. Remaining Gaps and Next Steps

Critical Fixes (Code Changes Required)

#	Fix	File	Priority
1	Behavioral Activation: Emerging → Strong	keystonePractices.ts	IMMEDIATE
2	Stage ¾ Expert: 30% → 38%	stageFramework.ts	IMMEDIATE
3	Remap 19 keystone practice lifts arrays	keystonePractices.ts	HIGH
4	Add Meta-Cognitive + Volitional to ALTITUDE_LINES	matchEngine.ts	MEDIUM
5	Revise Humor/Adversarial indep flags	axisModes.ts	MEDIUM
6	Replace scrambled mechanism texts	therapy-line mapping	MEDIUM

Research Still Needed (Perplexity Pass 2)

#	Gap	Prompts Available In
1	Mathematical line interventions	Loose Ends PDF, Prompt 2.1
2	Intuitive line interventions	Loose Ends PDF, Prompt 2.4
3	Aesthetic line interventions	Loose Ends PDF, Prompt 2.5
4	Adversarial line interventions	Loose Ends PDF, Prompt 2.6
5	Influence line interventions	Loose Ends PDF, Prompt 2.7
6	Matching algorithm validation	Loose Ends PDF, Prompts L2.1-L2.2
7	g-independence correlations for all 32 lines	Independence PDF, Block B
8	Discriminant validity for overlapping lines	Loose Ends PDF, Prompts L4.1-L4.4
9	Rarity composite statistical validity	Loose Ends PDF, Prompts L10.1-L10.2
10	Bottleneck roles empirical grounding	Loose Ends PDF, Prompts L7.1-L7.2

Appendix: All Verified DOIs (Quick Reference)

#	DOI	What It Proves
1	10.1207/s15326985ep3901_2	Philosophical/epistemic cognition develops through adulthood independently of IQ
2	10.1371/journal.pone.0238110	DIT-2 three-factor structure validated via CFA
3	10.1037/0021-9010.85.2.250	Action Control Scale (Tactical) is distinct from cognitive ability and personality
4	10.5334/spo.37	ACS-90 French validation confirms discriminant validity from Conscientiousness
5	10.3389/fpsyg.2015.01669	CPS (Architectural) has incremental validity over fluid intelligence
6	10.1027/2698-1866/a000009	Reduced MicroDYN battery shows excellent factorial validity
7	10.1371/journal.pone.0176218	PRFQ (Parenting) predicts infant attachment; three-factor structure validated
8	10.1177/2158244019867857	Mating Intelligence Scale CFA shows CFI 0.94-0.96
9	10.1037/t33090-000	SCI-2 (Community-Founding) alpha = 0.94, four-factor structure
10	10.1037/a0028446	Spatial training meta-analysis: g = 0.47, transfers, durable
11	10.3389/fnins.2021.630829	Adult music training causes structural brain reorganization
12	10.1186/S12877-020-01761-Y	5-week music training induces audio-motor plasticity in elderly
13	10.1257/jep.37.4.137	Financial literacy interventions cause modest but real gains
14	10.1038/s41598-019-44097-3	Nature 120-min threshold (observational, N=20,000)
15	10.1089/eco.2023.0063	Brief nature exposure produces short-term mental health benefits

#	DOI	What It Proves
16	10.1080/17439760.2025.2502483	Gratitude dosage comparison: letters/texts reach g = 0.40-0.60
17	10.1111/cpsp.12224	Expressive writing: no significant long-term effect on depression

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This document is the living master reference. Every time new research is verified, add it here. Every time a rating changes, update it here. Every time a DOI is confirmed or debunked, note it here. This is the empirical backbone of AQAL.

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9. Perplexity Pass 2: Independence from G (August 6, 2026)

Critical Findings That Change the Platform

Finding 1: Scale Back Independence Claims to 6-9 Lines

The original handoff targeted 12-15 genuinely independent lines. After rigorous verification, the defensible number is **6-9 lines** with credible independence from g. This is still enough to argue the platform measures something beyond IQ and personality — just not as much as originally hoped.

Finding 2: Volitional Is the Strongest Independence Case

Duckworth & Seligman (2005) found self-discipline outperforms IQ almost 2-to-1 in predicting academic outcomes ($r = 0.67$ vs $r = 0.32$), holding even after controlling for IQ directly. This is the single strongest non-g finding in the entire research corpus and the flagship empirical claim for the platform.

- **Citation:** Duckworth, A. L., & Seligman, M. E. P. (2005). Self-discipline outdoes IQ in predicting academic performance of adolescents. Psychological Science, 16(12), 939-944.
- **DOI:** 10.1111/j.1467-9280.2005.01641.x

Finding 3: RETRACT the Racetrack Handicapper Citation

Ceci & Liker (1986) “A day at the races” was directly rebutted by Detterman & Spry, who reanalyzed the same data and found the original DOES show an IQ relationship when correctly analyzed. This is a landmine claim.

- **Platform action:** REMOVE from all materials. Do not cite as evidence for practical intelligence independence.

Finding 4: Musical Intelligence is NOT Fully Independent of g

Burgoyne et al.'s twin study found music aptitude shares a genetic factor with g while retaining a unique genetic component. Musical is partially, not fully, independent.

- **Platform action:** Downgrade from "Unassailable" to "Partially Independent."

Finding 5: Adversarial/Machiavellian Has NO Validated Instrument

Strong evolutionary theory (Byrne & Whiten 1988) but literally no validated individual-differences psychometric instrument exists. Cannot be scored in rarity composite.

- **Platform action:** Flag as "theoretical construct without validated measurement." Exclude from rarity composite.

Finding 6: Practical Intelligence (Sternberg) is Contested

Gottfredson (2003) reanalyzed Sternberg's data and found r = 0.34 correlations with IQ. Not debunked but not independent either.

- **Platform action:** Present both sides. Do not claim full independence.

Finding 7: Interoceptive Accuracy Holds Up (With Measurement Caveat)

Low g-correlation confirmed, but heartbeat-counting method has known validity problems (Zamariola et al. 2018).

- **Platform action:** Keep as independent but note measurement caveat.

Revised Independence Scorecard

Line	Independence Status	Key Evidence	Confidence
Volitional	INDEPENDENT	Duckworth 2005: r=.67 vs IQ r=.32	HIGH
Interoceptive	INDEPENDENT (with caveat)	Low g-correlation; measurement issues	MEDIUM-HIGH
Kinesthetic	INDEPENDENT	Motor factor (Gm) in CHC; separate from Gf	HIGH
Empathic	PARTIALLY INDEPENDENT	MSCEIT r~0.30 with IQ; 85%+ unique variance	MEDIUM-HIGH
Resilient	PARTIALLY INDEPENDENT	CD-RISC not predicted by IQ	MEDIUM
Interpersonal	PARTIALLY INDEPENDENT	Social intelligence low g-loading	MEDIUM
Musical	PARTIALLY INDEPENDENT (downgraded)	Shares genetic factor with g	MEDIUM
Aesthetic	PARTIALLY INDEPENDENT	Openness-based; not cognitive	MEDIUM
Adversarial	THEORETICAL ONLY	No validated instrument	LOW
Spatial	G-LOADED (CHC: Gv)	r~0.60 with g	NOT INDEPENDENT
Logical	G-LOADED (IS Gf)	Basically IS fluid intelligence	NOT INDEPENDENT
Mathematical	G-LOADED (CHC: Gq)	High g-correlation	NOT INDEPENDENT
Linguistic	G-LOADED (CHC: Gc)	r~0.70 with g	NOT INDEPENDENT

Defensible independent lines: 6-9

Platform Actions Required

1. Adjust rarity composite to weight independent lines more heavily
2. Commission factor-analytic study when 500+ members assessed
3. RETRACT racetrack handicapper citation
4. Present independence claim honestly: “6-9 dimensions beyond g”
5. Flag Adversarial as unmeasurable until instrument exists
6. Downgrade Musical from fully to partially independent